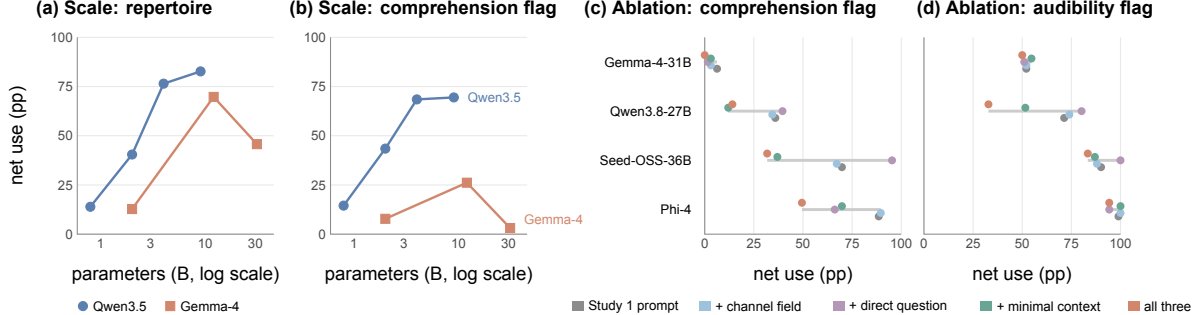
(a) Scale: repertoire
(b) Scale: comprehension flag
(c) Ablation: comprehension flag
(d) Ablation: audibility flag
net use (pp)
100
75
50
25
0
1
3
10
30
parameters (B, log scale)
Qwen3.5
Gemma-4
Gemma-4-31B
Qwen3.8-27B
Seed-OSS-36B
Phi-4
0
25
50
75
100
net use (pp)
Study 1 prompt
+ channel field
+ direct question
+ minimal context
all three